

# Chunbo Li

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## EXPERIENCE

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| <b>Xiaomi</b>                                                                                                                       | <b>Sep. 2023-Present</b>   |
| <ul style="list-style-type: none"><li>Software Development Engineer / Focus on Performance Optimization of Android System</li></ul> |                            |
| <b>Nanyang Technological University</b>                                                                                             | <b>Aug. 2021-Dec. 2023</b> |
| <ul style="list-style-type: none"><li>Master of Engineering / Major in Computer Science</li></ul>                                   |                            |
| <b>Tongji University</b>                                                                                                            | <b>Sep. 2017-Jun. 2021</b> |
| <ul style="list-style-type: none"><li>Bachelor of Engineering / Major in Computer Science</li></ul>                                 |                            |

## ACADEMIC PAPER

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| <ul style="list-style-type: none"><li><b>D&amp;A: Resource Optimisation in Personalised PageRank Computations Using Multi-Core Machines</b></li><li><b>TKDE Conference, 2nd Author</b></li></ul>               | <b>Jun. 2024</b> |
| <ul style="list-style-type: none"><li><b>Personalized PageRanks over Dynamic Graphs -- The Case for Optimizing Quality of Service</b></li><li><b>ICDE Conference, 6th Author</b></li></ul>                     | <b>Dec. 2023</b> |
| <ul style="list-style-type: none"><li><b>[Experiments &amp; Analysis] Multi-Task Processing in Vertex-Centric Graph Systems: Evaluations and Insights</b></li><li><b>EDBT Conference, 5th Author</b></li></ul> | <b>Jan. 2023</b> |

## RESEARCH EXPERIENCE

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| <b>Response Time Optimized Personalized PageRank on Multi-Cores System</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Oct. 2021-Aug. 2023</b> |
| <ul style="list-style-type: none"><li>I proposed the <b>Parallel-Friendly Shuffling method</b> to handle massive PageRank queries in dynamic graph, and obtained a reduction of both response time and processing time of SSPPR queries.</li><li>Proved that <b>graph updates may only cause limited inaccuracy in queries</b>, showing that queries can be processed behind previous updates with higher efficiency.</li><li>Proposed <b>parallel PPR processing algorithm</b> to handle queries on dynamic graph.</li><li>Conducted extensive experiments on datasets with large size, <b>achieving at most 5.76 speedup</b>, comparing to baseline models.</li><li>Co-author of <i>Personalized PageRanks over Dynamic Graphs -- The Case for Optimizing Quality of Service</i>.</li></ul> |                            |
| <b>Resource Optimization in Personalized PageRank Computations Using Multi-Core Machines</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Aug. 2021-May. 2022</b> |
| <ul style="list-style-type: none"><li>We proposed D&amp;A, which <b>determines the optimal number of processing threads</b> needed to complete a workload within a given time constraint.</li><li>I <b>designed an efficient parallel framework</b> to conduct experiments and validation.</li><li>We achieved reductions in the required number of threads ranging from 38.89% to 73.68%.</li><li>Co-author of <i>D&amp;A: Resource Optimisation in Personalised PageRank Computations Using Multi-Core Machines</i>.</li></ul>                                                                                                                                                                                                                                                              |                            |
| <b>Multi-Task Processing in Vertex-Centric Graph Systems: Evaluations and Insights</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Aug. 2021-May. 2022</b> |
| <ul style="list-style-type: none"><li>This research presented <b>several important insights</b> on the correlation between system performance and configurations.</li><li>We also proposed a <b>cost-based tuning framework</b> that optimizes the performance of a representative VC system.</li><li>I contributed to the <b>experiment design and data processing</b>.</li><li>Co-author of <i>[Experiments &amp; Analysis] Multi-Task Processing in Vertex-Centric Graph Systems: Evaluations and Insights</i>.</li></ul>                                                                                                                                                                                                                                                                  |                            |
| <b>Earlier Research Projects on Data Mining</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                            |
| <ul style="list-style-type: none"><li><b>Stock Market Predictions:</b> Utilized the BERT model to obtain sentimental information for prediction.</li><li><b>Traffic Congestion Prediction:</b> Researched the potential correlation between the roads.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                            |

## WORK EXPERIENCE

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| <b>Xiaomi Corporation</b>                                                                                                                                                                                                                       | <b>Aug. 2023-Present</b> |
| <ul style="list-style-type: none"><li>Our team developed the <b>Performance Optimization Service</b> in smart cockpit system, to enhance system robustness and efficiency, and <b>my work focuses on optimizing the CPU schedule</b>.</li></ul> |                          |

## Scheduler Optimization in Linux Kernel

- Researched the Windows-Assist Load Tracing (WALT) algorithm.
- We **refined the load-balancing algorithm** to relieve the overload problem.
- Co-designed a **specialized "top" tool** based on WALT to monitor CPU load, using a normalization method.

## Scheduler Policy Development in User Level

- To monitor the schedule of process activities, we designed the **schedule boost policy** for adjusting application priority and CPU affinity, and **VIP queue** policy for computing power allocation with higher accuracy.
- We further let the VIP scheduler to **be enabled only when touch events occur**, to enhance schedule more precisely.
- By monitoring the status of processes, the boost policy is applied to **improve the frames per second, start time, and response time**.

## IO Performance Monitor

- To solve the janky problems caused by IO wait times (iowait), we **monitor the IO performance** according to PSI value.
- The **processes with high IO rate and fsync times** are recorded when PSI reach the threshold.

## System Performance Analysis

- When performance-related problems occur, we **analysis the system log** and find out the reason.
- Apply optimization method to solve the problems, improving the robustness of the system.

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## RESEARCH INTERESTS

Operating System, Resource Allocation, System Tuning, Parallel Algorithm

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## AWARDS

### Xiaomi

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| Coding Master (Nominated)                                          | Jan. 2025 |
| “The Climb” Award to Our Team for Solving Hard Debugging Challenge | Jan. 2024 |

### Tongji University

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| Second Prize in Shanghai Division of Contemporary Undergraduate Mathematical Contest in Modeling | Sep. 2020 |
| Third Prize of Mathematical Modeling Competition granted by Tongji University                    | May. 2019 |
| Third Class Scholarship granted by Tongji University                                             | Dec. 2018 |
| Second Prize of Programming Contest granted by Tongji University                                 | May. 2018 |

### Senior High School

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| Provincial First-Class Prize of National Olympiad in Informatics in Provinces | Feb. 2017 |
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## SKILLS

### Coding Skills

- Programming Languages: C++ (Practised), Java (Practised), Python (Good)

### Expertise

- eBPF, Linux